

Maintenance sheet version 9 for ablation area (tripod, one level)

Station name	TAS_L
People present	AOEP + ASO + PILOT
Purpose	Maintenance / installation / take-down / quick visit / emergency
Date	3/9-22 / 520

Notes:

Station tilted over on arrival, CNR and wind sensor destroyed. Will replace with parts from TAS-A

AWS to do list		Check after maintenance			
Part	Shipped (Y/N)	Replaced? (Y/N)	Number shipped/arrival	Number departure	
Radiometer	N	Y (old from Tas A)		171267	Clean? (Y) / N
Ablation hose assembly	N	N		5087	No air in hose and bladder half full? (Y) / N
Iridium modem IMEI number	N	~		300234010884770	
Inclinometer/compass	N	N	Attached to the radiometer? Y / N		
Satellite antenna	N	N			
Wind sensor	N	Y old from Tas A	Black box pointing toward the radiometer (i.e., along the boom)? Y / N		
Temperature / humidity assembly (Rotronic)	Y	Y	Fan spins? Y / N Sounds, OK? Y / N		
HygroClip T/hum assembly	N		NB: Rotronics (Y / N) could have a 40 °C offset. New or old type? NB: Luft sensor (Y / N)		
Sonic ranger on AWS	Y	Y	Old sensor with new membrane? Y / (N)		
Sonic ranger on stake(s)	Y	Y	Old sensor with new membrane? Y / (N) NB: make sure not being buried next winter		
Thermistor string	Y	Redrill the one that melted out	Same sensor, 3 sensors in ice. (Drill got stuck)		
Solar panel	N	N	Clean? (Y) / N NB: is not effective when partially covered		
GPS antenna	N	N	At top of logger enclosure? (Y) / N		
Data logger	N	N	CR1000 or CR1000X?		
Multiplexer	N	N	Clicking during measurement cycle? (Y) / N		
Logger enclosure	N	N	Replaced (including everything inside)? Y / (N) New desiccant bags? (Y) / N		
Battery box	N	N	New box? Y / (N) New desiccant bags? (Y) / (N)		
Stakes	Y	Y	New one(s) drilled in? Y / N Length of new stake(s) 8 m? Y / N Bottom plugged to reduce melt-in? Y / N		
Tripod	N	N (Y)	New tripod? Y / (N) → Bent during fall, got stuck New sensor boom? (Y) / N		

Metadata maintenance

	Arrival	Departure
Time difference between logger clock and UTC ($\pm$ mm:ss)		
Download data and/or change CF/SD Card (wait for green light)	Y / N	Y / N
Name of logger program		
Screen dump of values in fast scan mode	Y / N	☒ Y / N
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N	Y / N
Mast tilt towards radiometer (- if tilting towards anemometer)	$^{\circ}$	-2°
Mast tilt across boom looking from radiometer (- if clockwise)	$\times$ $^{\circ}$	2°
Boom direction relative to north	$^{\circ}$ true/magn.	☐ $^{\circ}$ true/magn.
Box on wind sensor exactly towards mast? If not, measure.	Y / N	☒ Y / N
Radiometer exactly aligned with inclinometer? If not, measure.	Y / N	☒ Y / N
Temperature / humidity height (bottom of casing – surface)	cm	262 cm
Stake assembly standing? Broken	Y / ☒ N / sort of	☒ Y / N / sort of
Free length of stake (if 3: outer, holding boom with sonic ranger)	cm	cm
Free length of stake 2 (if 3: middle)	cm	149 cm
Free length of stake 3 (if 3: outer, without sonic ranger)	cm	cm
Height of ablation bladder fluid level above the ice surface	cm	137 cm
Vertical difference in surface height (at station – at hose) (approx.)	cm	-20 cm
Length of thermistor string on surface from surface marking	cm	900 cm
Apparent damage to sensors? If yes, which?	Y / N	Wind sensor ☒ Y / N

Notes:

- x Boom replaced
- x CNR from rns-A used
- x WS & D from rns-A used
- x CF stance broken on arrival -> replaced

mill has a crack
(see photos)

Must-do at next visit:

- x New string
- x New WS & D
- x New CNR (old one is from 2019)